School of Computer Science, University of Bristol

COMS30087: Image Processing and Computer Vision, AY: 2026/27

Assignment: 3-D from Stereo

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This assignment assumes that you have completed the two stereo lab sheets. It involves designing and implementing algorithms to determine corresponding spheres in two views of a scene consisting of spheres on a plane and using the correspondences to reconstruct the spheres, their centre points and radii. You are required to submit a report and your code. The aim is to demonstrate your ability to convert theory into practical algorithms and to write a report that motivates, explains and justifies your method and presents an evaluation of the method and an analysis of the results.

Important: . Both the report and the code are required to obtain marks. If you submit just code and no report, or vice versa, you will be awarded zero marks. They should be complementary and consistent - the report should reference the code and the code must reference and be consistent with what is written in the report. Your mark will be based on the content of the report and the code.

Important: There are a number of ways to tackle this assignment. However, you must use the methods described in the lectures. If you use any alternative methods, then zero marks will be given. If you refer to theory or techniques in the report that are not covered in the lectures, then their relationship with the material that is covered must be clearly explained, otherwise the relevant section will not be used for assessment. Your submission must be your own work and the use of AI for either the code generation or the report is not allowed.

Assignment Tasks

1. Download and run the assignment version of the 3-D simulator used in lab sessions. Refer to Lab Sheet I for details.

2. You should see 6 spheres of different sizes located on a plane. You will find two images in your directory corresponding to the images captured by two virtual cameras (VCs). Each time you run the code you will see a different arrangement of spheres and images captured from different viewpoints. You can change the number, separation and size of the spheres and show the sphere centres via command line arguments. Experiment with the different parameters to see the impact on the VC images.

3. Use the openCV function cv2.HoughCircles() to detect circles in each VC image. Check the results for several examples, noting any dependencies on any parameter settings.

4. Select one of the VCs as the reference view and for each detected circle centre in its image, compute the corresponding epipolar line for the other image. Draw the line and check that it is correct.

5. Use the epipolar line for each detected circle centre in the reference view to find the corresponding circle in the other view.

6. For each circle correspondence, compute the 3-D location of the associated sphere centre using the 3-D reconstruction algorithm described in the lectures.

7. Display the estimated sphere centres alongside the ground-truth centres in the visualisation. Compute the errors in the sphere centre estimates.

8. Using the circle detections in the reference and viewing images, estimate the radius of each sphere.

9. Display the estimated spheres alongside the ground truth spheres and compute the error in the radius estimates. Think carefully about how you display the estimated and ground truth spheres so as to allow comparison, since comparison may be difficult if you display the estimates as solid spheres.

10. Investigate the performance of your implementation using different sphere sizes and separations. Note when it does work and when it fails, and find out why in the latter case.

Note: For some of the above tasks you may wish to change the code so that the number and configuration of spheres gives a specific scenario rather than being random. This is fine, but you should add comments accordingly and comment out the code before submission so that random generation is restored.

Submission

1. **Code** Your code must be commented. You must insert comments that explain the implementation of key components and makes explicit reference to sections and equations in your report. Marks will be deducted if these explanations are not included. Your code will be tested and the comments checked. You must provide a clear explanation of how to run the code at the top of the source file.

2. **Report** Your report should be no more than 8 single column pages or 4 double column pages. Pages beyond these limits will not be marked. The text area size should be the same as this document and you should use a similar font (11pt). At the top of the first page, put the title 'COMS30087 IPCV CW 2026-27', your username and your student number.

   The report should consist of the sections listed below. For the method sections, give full details of your implementation and evaluation, describing and justifying the methods and any assumptions, and providing an analysis of the results. **You must reference relevant slides in the lectures when giving details of your algorithm and method**. In addition, each section should include the content indicated below.

   (a) **Introduction** A brief introduction outlining the assignment, the objectives, what you achieved and what you discovered.

   (b) **Epipolar lines and correspondence**. Provide an example of successful matching and an example of when it fails, showing the epipolar lines and the correspondence estimates.

   (c) **Sphere centre estimation**. Provide one example of estimated sphere centres after successful correspondence matching and one example when mismatches have occurred. Show the ground truth and estimated centres and provide a table listing the ground truth, estimated locations and the error. You must explain the data in the table, including the definition of how you computed the error. You should also explain the impact of the mismatched cases.

   (d) **Sphere radius estimation**. Provide one example showing good estimates and one where the method fails, including the ground truth and estimated values in each case.

   (e) **Noisy pose**. Investigate what happens if you use noisy relative pose between the cameras (position and orientation), rather than the ground-truth. Show examples of the impact on the correspondence matching and on the estimation of the sphere centres and radii. Compare the errors obtained wrt the level of noise in the pose. You must give details of how you added noise to the pose and justify the method.

Marking Scheme

Marks will be awarded based on your report and your code **in combination**. The method sections (b)-(d) will be marked out of 20 and an additional 20 marks will be awarded for the overall style and content of the report. To gain full marks for a section, you need to have completed all of the tasks, submitted valid code that works, and provided descriptions and examples in the report that provide strong evidence that you have fully understood the tasks and results.

A note on the report style. It will be useful if you regard your report as a mini textbook on stereo matching and reconstruction. The target reader might be a fellow CS student not taking the unit. Your report should be as self contained as possible and provide them with clear descriptions and explanations of the theory and implementation, and an indication of how well the methods work. Make sure you clearly define and explain all terms and make good use of equations and figures where appropriate.